

Avinash Nayal

Norman, OK | avinash.nayal@gmail.com | 4807658044 | Google Scholar

SUMMARY

Ph.D. candidate in Mechanical Engineering with five years of experience integrating mechanical design, thermal systems, and materials engineering. Designed end-to-end design of tooling and assemblies, including Couette-shear alignment system for high-k polymer films, using SolidWorks, GD&T, and DFM/DFA principles. Developed FEA/ANSYS and Python/MATLAB models to guide design decisions. Revamped lab concepts into manufacturable thermal builds, delivering $\Delta T < 2^\circ\text{C}$ in battery thermal modules and $>30\times$ thermal-conductivity gains in phase-change composites.

EDUCATION

University of Oklahoma | Norman, OK - Ph.D. in Mechanical Engineering

Aug 2020 – Present

Arizona State University | Tempe, AZ - M.Sc. in Mechanical Engineering

Aug 2018 – Jul 2020

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant - Phase 2 | University of Oklahoma

Jan 2023 – Present

Micro/Nano Heat Transfer Lab | *Ph.D. Advisor:* Dr. Jivtesh Garg

- Designed a Couette-shear extrusion tool under constraints on concentricity, sealing, chemical compatibility, minus 80°C transfer, and serviceability.
- Released GD&T drawings, structured BOM, and SOPs for priming, purge, RPM ramp, nozzle-to-mold alignment, and teardown to standardize builds.
- Applied DFM/DFA/DFC to modularize subassemblies, reduce fasteners, and specify finishes/fixtures so the shop could hold tolerances without blowing cost.
- Led RCA and commissioning; resolved air ingress, seal weep, and clog issues, then defined a run window and acceptance tests for repeatable laydown.
- Collaborated on comparative testing of air, liquid, CPCM, and hybrid cooling methods for battery thermal management, achieving uniform temperature control ($\Delta T < 2^\circ\text{C}$).

Graduate Research at ORNL (GRO) Intern | Oak Ridge National Laboratory (ORNL)

Jan 2025 – May 2025

Materials Science and Technology Division | *Mentor:* Dr. Lucas Lindsay

- Coordinated across physicists, materials scientists, and computational engineers to structure modeling workflows and deliver engineering-grade results under tight timelines.
- Took ownership of a stalled phonon-analysis workflow, rebuilt the pipeline end-to-end, and improved reliability, documentation, and data traceability.
- Developed a Python-based post-processing framework that standardized file structures and analysis routines, cutting cycle time by 40% and making the workflow plug-and-play for the team.
- Learned new tools (Fortran, HPC workflows, QE/Phonopy) to resolve data failures and produce clear engineering documentation, reproducibility notes, and high-quality visualizations.

Graduate Research Assistant - Phase 1 | University of Oklahoma

Aug 2020 – Dec 2022

- Engineered and characterized expanded-graphite (EG) composites with a 3070% increase in thermal conductivity for advanced heat-spreader applications.
- Optimized graphene-epoxy thermal interface materials by tailoring solvent selection (DMF), achieving 44% higher thermal conductivity and bettered nanoparticle dispersion.
- Computed thermal conductivity ($162\text{ Wm}^{-1}\text{ K}^{-1}$) of hexagonal BC_2P , via first-principles modeling.
- Mentored 2 junior researchers on thermal testing and data analysis, standardizing Raman/DSC/TGA characterization workflows for materials testing and improving data consistency across experiments.

Master's Thesis | Arizona State University

May 2019 – Jul 2020

Nano-Engineered Thermal Radiation Group | *Ph.D. Advisor:* Dr. Liping Wang

- Devised a coupled radiation-conduction-electrical MATLAB model for solar thermophotovoltaic (STPV) systems, validated against experimental results with $< 5\%$ deviation.
- Optimized multilayer absorber/emitter metafilms using Particle Swarm Optimization and fabricated coatings via PECVD-sputtering, achieving high spectral selectivity ($0.4 - 1.6\text{ }\mu\text{m}$).
- Engineered a vacuum-based STPV test setup with Xe-arc solar simulator and SiO microsphere spacers to maintain a $100\text{ }\mu\text{m}$ gap, demonstrating negligible conduction loss and enhanced radiative efficiency.

Research Intern | Indian Institute of Technology Gandhinagar

Aug 2017 – Jul 2018

Mechanical Engineering Department

- Led testing and overhaul of a 5 kWe diesel-powered fuel-cell reformer system (DRDO-funded).
- Developed scale-up plans from 5 kWe to 30 kWe, integrating ASPEN modeling, mass/energy balances, and PID control requirements to support DRDO's prototype roadmap.
- Created updated P&IDs and mechanical layouts for heat-exchanger, piping, and DAQ/control subsystems, enabling a more modular and scalable test rig.
- Coordinated with fabrication vendors and internal teams to define component specifications, procurement lists, and tenders for high-value items, ensuring manufacturability and compatibility.

Department of Atomic Energy | *Advisor:* Dr. Vinay Menon

- Simulated conjugate heat transfer and coolant flow in the ITER divertor under $> 10 \text{ MW/m}^2$ heat flux using ANSYS APDL, evaluating swirl-tube vs. straight-flow geometries.
 - Validated simulation results with calorimetry and RTD-based measurements, recommending design improvements that achieved 30% better thermal uniformity in the ITER divertor cooling system.
 - Derived empirical peaking-factor correlations capturing heat-flux non-uniformity in high-heat-flux liquid-cooling channels, providing predictive inputs for divertor design optimization within $\approx 10 \%$ of experimental data.
-

SKILLS

CAD & Design: SolidWorks, Fusion 360, TINKERCAD

Simulation & Analysis: ANSYS(Fluent, APDL), COMSOL, Thermal/CFD Modeling, First-Principle calculations, FEA.

Programming & Data: Python(workflow automation, numerical modeling, data processing), Matlab, C++

Materials & Testing: Differential Scanning Calorimetry (DSC), Thermo-gravimetric Analysis (TGA) Raman Spectroscopy, XRD, SEM, Thermal conductivity testing.